

Xiangjian Zeng

azeng@mit.edu | (412)579-1999 | www.linkedin.com/in/xiangjian-zeng | aaronzeng-cmu.github.io/

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Ph.D. in Electrical Engineering and Computer Science

Expected Jun 2029

GPA: 5.00/5.00

Carnegie Mellon University, Pittsburgh, PA

M.S. in Electrical and Computer Engineering

Dec 2023

GPA: 4.00/4.00

University of California San Diego, La Jolla, CA

B.S. in Electrical and Computer Engineering (Minor in Cognitive Science)

Dec 2020

GPA: 3.91/4.00, Magna Cum Laude

RESEARCH EXPERIENCE

Lewis Lab, Massachusetts Institute of Technology, Cambridge, MA

09/2024-present

Closed-loop Auditory Modulation of Cerebrospinal Fluid during Sleep

- Develop real-time EEG processing and machine-learning-based slow-oscillation phase-prediction algorithms for precisely timed closed-loop auditory stimulation during sleep
- Integrate simultaneous EEG-fMRI with electrophysiological and CSF-flow measurements to quantify stimulation-evoked neural and physiological dynamics across younger and older adults
- Co-authored an accepted Science Translational Medicine study demonstrating that closed-loop auditory stimulation targeting sleep slow waves modulates cerebrospinal fluid flow in humans, motivating development of personalized stimulation algorithms

Robust Automatic Sleep Staging for Simultaneous EEG-fMRI

- Benchmark automatic sleep-staging models on manually staged simultaneous EEG-fMRI and characterize systematic failures under scanner-domain shift
- Investigate contributions from temporal/context priors, sensor and reference mismatch, and scanner-related signal differences using controlled model and preprocessing comparisons

Insanally Lab, University of Pittsburgh, Pittsburgh, PA

05/2023-08/2024

Bayesian Ensemble Neural Decoding

- Developed a probabilistic ensemble neural decoder, combining inter-spike-interval-based conditional probabilities across multiple neurons to jointly predict behavioral trial conditions, revealing that traditionally neglected “quiet” non-classically responsive neurons interact with responsive neurons in learning
- Demonstrated that aggregating non-classically responsive population activity increased behavioral decoding accuracy from 60% to 65% during late-stage learning compared to early, revealing information not captured by classical response-selective neurons

Biologically Constrained Spiking Recurrent Neural Networks

- Built 1,000-unit spiking recurrent neural-network models in Python and Julia under biological constraints to simulate cortical dynamics underlying auditory discrimination
- Investigated how flexible network dynamics emerge from balanced excitatory and inhibitory activity and developed a computational model describing the resulting dynamics

Chamanzar Lab, Carnegie Mellon University, Pittsburgh, PA

05/2022-04/2023

Neural Probe Waveguide Characterization System

- Developed an automated neural-probe waveguide characterization pipeline integrating computer vision, robotics, and signal processing, enabling non-destructive loss measurement previously achievable primarily through destructive methods; first-authored a peer-reviewed conference proceedings paper

Adaptive Neural Signal Noise Cancellation

- Developed and hardware-validated an adaptive filtering algorithm for peripheral nerve recordings, achieving approximately 45 dB suppression of 60-Hz interference; presented the work as first author at the North American Neuromodulation Society meeting
-

SELECTED MACHINE LEARNING PROJECT

Deep Learning Automatic Fake News Detection

Fall 2022

- Developed a BERT-based language model to retrieve relevant evidence and classify misinformation in political statements, achieving 72% binary classification accuracy
-

SELECTED PUBLICATIONS & PRESENTATIONS

- Levitt, J., **Zeng, X.**, Yang, Z., Jacob, L. P. L., Lewis, L. D. “Closed-loop auditory stimulation targeting sleep slow waves drives cerebrospinal fluid flow in humans.” *Science Translational Medicine*. Accepted, 2026.
 - **Zeng, X.**, Reddy, J., Chamanzar, M. “Automated Photonic Waveguide Loss Measurement using Out-scattering Light Method.” *Integrated Systems Design and Technology Conference Proceedings*, 2023.
 - Toth, J., Sidleck, B., Lombardi, O., et al., **Zeng, X.**, et al. “Dynamic gating of perceptual flexibility by non-classically responsive neurons.” Manuscript under review.
 - **Zeng, X.**, Yu, G., Albana, B. F., Insanally, M. N. “Contributions and synaptic basis of diverse cortical neuron response to flexible task performance.” *COSYNE*, 2024.
-

FELLOWSHIP

- MIT EECS MathWorks Fellowship 2026
 - MIT Artificial Intelligence and Wellness Fellowship 2025
-

TECHNICAL SKILLS

- Programming: Python, C/C++, MATLAB, Julia, R
- Machine Learning & Modeling: PyTorch, TensorFlow, scikit-learn | neural decoding, deep learning, spiking neural networks, computational modeling
- Signal & Multimodal Data: EEG, EEG-fMRI, electrophysiology, adaptive filtering, real-time signal processing